Research Design: Home Run Derby Participation and Change in Batted-Ball Approach

This design builds on the domain and skill review by narrowing the research question from performance to approach. Performance statistics such as OPS and WAR are shaped by pitching, defense, and luck as well as by a hitter's own swing, so a real change in approach can be masked or exaggerated over a second half of roughly 65 to 70 games. Batted-ball data provides a more direct description of the hitter's approach. The research question is: how, if at all, does a Home Run Derby participant's batted-ball profile change from the first half to the second half of the season, relative to a matched group of non-participating All-Stars?

The cause under study is Derby participation; the proposed effect is a shift toward a more pull-side, air-ball-dominant profile. Derby success rewards pulling the ball in the air, as it is easiest to hit home runs near the foul line, where the fence is shallower and thus a home run does not need to travel as far. Creating a pull-side air fly ball requires contact earlier in the pitch's flight, and Giordano et al. (2026) found contact depth closely tied to swing attack angle, while Nakashima et al. (2025) showed the acceptable timing window at impact is only several milliseconds and depends on swing path. Repeated maximal-effort, pull-air swings could plausibly leave a trace in a player's regular swing timing, which this design tests for directly rather than assuming.

Purpose and Causal Logic

This is a non-experimental study: a player is invited to the Derby based on first-half home run totals and chooses whether to accept, so participation is self-selected on exactly the kind of hitter who already favors an extreme, pull-air approach (hitters who routinely pull the ball in the air with high exit velocity will naturally have high home run totals). The design can at most show whether participants' batted-ball profiles shifted more than a matched comparison

group's, after accounting for first-half level to net out ordinary regression to the mean (Barnett, van der Pols, & Dobson, 2005). It cannot rule out unmeasured factors correlated with both Derby selection and a swing change (a coaching adjustment made for unrelated reasons, say), and it cannot generalize beyond the seasons and formats studied. In short, the design supports a claim of association under a controlled comparison, not causation in the experimental sense.

The comparison group pairs each participant with a non-participating All-Star of similar first-half swing profile, a quasi-experimental matched design in the spirit of DeFroda et al. (2021) and Sobol et al. (2024), matched on swing approach rather than home run total. If too few comparable non-participants exist within a cluster, the backup design falls back to a within-subject comparison following Jaiclin and McCollum (2010): each participant's Derby-year shift against their own shift in seasons without participation, which removes between-player confounds but requires multiple seasons per player and reintroduces era-level differences (the 2026 format change, say) as a confound of its own.

Data

Data will be drawn from three public sources, summarized in Table 1: Baseball Savant, FanGraphs, and Baseball-Reference, all freely browsable and used routinely in peer-reviewed baseball analytics work (Koseler & Stephan, 2017). Savant's data is made available by MLB for public research and analysis; FanGraphs and Baseball-Reference permit non-commercial academic use with attribution. No data will be resold or redistributed, and no permission beyond attribution is required.

One variable will be hand-coded rather than pulled directly: home-run-specific spray direction, split by half-season, since Savant exports overall batted-ball direction only as a season aggregate. Each home run will be classified as pull, middle, or opposite field from the player's

Statcast spray chart (handedness defines "pull") using a written codebook fixed before coding begins, with about 10% re-coded blind afterward to check consistency, since this is the one step depending on human judgment.

Source	Primary variables pulled	Use policy
Baseball Savant	Exit velocity, launch angle, pull/oppo%, barrel%, bat speed/attack angle (2023+), HR spray charts	Public research/commentary use
FanGraphs	Split-season batted-ball direction/trajectory (Pull%, Cent%, Oppo%, GB%/FB%/LD%) via the Splits Leaderboard's 1st/2nd Half preset; plate appearances	Non-commercial academic use, attribution
Baseball-Reference	Season schedule, All-Star break date, All-Star rosters, split stats	Non-commercial academic use, attribution

The three sources use three different player identifiers (an MLBAM ID on Savant, a separate FanGraphs ID, a separate Baseball-Reference key), merged via a crosswalk (the Chadwick Bureau register, through pybaseball) rather than by name, since names collide and are spelled inconsistently across sites. The data form a player-season-half panel: first half and second half nested within season, nested within player, with multiple seasons pooled once collection is complete. Inclusion criteria are a minimum plate-appearance total in both halves and position players only. Timer-based seasons (2015-2025) and the swing-based 2026 format are kept as separate regimes given the difference in swing volume, and bat-tracking variables (bat speed, attack angle) are restricted to 2023 onward, excluding earlier seasons from that part of the analysis only. All data describe public, on-field performance with no private information or institutional-review concerns; the main risks are the manual coding, addressed above, and which comparison players are available to match against, addressed by the clustering and backup-design logic above.

Statistical Analysis

The analysis proceeds in three steps. First, PCA will reduce the standardized batted-ball and swing-trait variables (pull%, air-ball%, launch-angle distribution, exit velocity, strikeout rate, and, where available, bat speed and attack angle) into a small number of components describing a hitter's overall approach, following Abdi and Williams (2010) and the retention guidance in Jolliffe and Cadima (2016). Hitters will then be grouped on their first-half component scores using k-means clustering, with the number of clusters chosen by the elbow method defined by Camp, Sarris, and Dvorocsik (2020). This clustering both defines the pull-air approach category used as a moderator, and supplies the pool from which each Derby participant's matched control is drawn.

Second, paired t-tests will test whether each player's first-half and second-half component scores (and individual rates such as strikeout rate and pull-air percentage) differ, and independent-samples t-tests will compare the size of that change between participants and their matched controls, with every difference reported as a 95% confidence interval rather than a p-value alone, since an interval on the size of the shift is more informative given a sample likely in the low hundreds.

Third, a multiple regression will predict each player's second-half component score from Derby participation, the first-half score (netting out regression to the mean, per Barnett et al., 2005, and Jaiclin & McCollum, 2010), the pull-air moderator, and their interaction, mirroring the multivariable approach DeFroda et al. (2021) used for performance outcomes, applied here to approach instead. Before interpretation, regression assumptions will be checked, cluster count validated with a silhouette score, and an a priori power calculation will establish what interaction

effect size the design can realistically detect, so a null result is judged against that benchmark rather than assumed to mean no effect exists.

References

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Note: References follow APA (7th edition) format.